Compressible Flow Validation in a Quasi-2D Laval Nozzle

OpenFOAM `rhoCentralFoam` Portfolio Report

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1 Abstract

This report summarizes a compact CFD validation project for ideal-gas air flowing through a quasi-2D converging-diverging Laval nozzle. The cases are configured, run, and validated with OpenFOAM `rhoCentralFoam`, `blockMesh`, shell wrappers, Makefile targets, and deterministic Python post-processing scripts. Current capabilities include pressure-ratio regime classification, direct mass conservation from saved ρ and $\mathbf{U}$ fields, Courant-number stability assessment, isentropic centerline diagnostics, quasi-1D area-Mach comparison, time-history extraction, and mesh-sensitivity reporting. The project is intended as a professional CFD portfolio artifact, not as a production-grade nozzle design workflow.

2 Project Overview

The model is a quasi-2D Laval nozzle with `empty` front/back patches and intentionally inviscid-style slip walls. Slip walls are used to support comparison against ideal isentropic and area-Mach relations rather than to represent viscous wall losses or boundary-layer development.

3 Physical Background

For ideal compressible nozzle flow, reducing the back-pressure ratio p_b/p_0 changes the flow topology. High back pressure keeps the nozzle fully subsonic. Near the critical pressure ratio for air, the throat approaches Mach 1 and the mass flow chokes. Lower back pressure permits

Table 1: Project configuration.

Item	Configuration
Solver	OpenFOAM `rhoCentralFoam`
Mesh generator	`blockMesh`
Geometry	quasi-2D converging-diverging Laval nozzle
Gas model	ideal-gas air
Reservoir pressure	$3.000\,00 \times 10^5\,\mathrm{Pa}$
Reservoir temperature	$3.00 \times 10^2\,\mathrm{K}$
Walls	`slip`, intentionally for inviscid/isentropic validation
Mass-flow method	direct $\int_A \rho\mathbf{U} \cdot \mathbf{n}\, dA$ from fields
Flux-field usage	`phi`, ρ`phi`, and `rhoPhi` not used

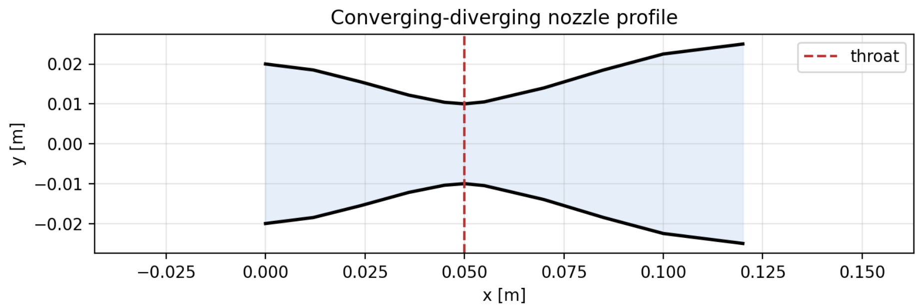

Figure 1: Quasi-2D Laval nozzle geometry used for the OpenFOAM cases.

Table 2: Pressure-ratio validation results from existing computed outputs.

Case	p_b/p_0	Throat M	Max M	Mass error	Verdict
`subsonic`	0.966667	0.480536	0.480536	0.690756%	valid
`choked`	0.528333	1.03652	2.30388	0.277965%	valid
`internal_shock`	0.466667	1.03577	2.36018	0.256732%	valid

supersonic acceleration downstream of the throat; if the imposed back pressure is above the shock-free supersonic exit pressure, the divergent section can contain a normal shock.

The inviscid area-Mach relation used for validation is

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}, \tag{1}$$

with $\gamma = 1.4$. The relation has subsonic and supersonic branches for $A/A^* > 1$. It is not applicable through a shock because entropy is produced and the effective downstream A^* changes.

4 Numerical Setup

The cases use `rhoCentralFoam`, an explicit density-based compressible solver. Meshes are generated with `blockMesh`; the baseline and medium mesh contain 70000 cells. The computed fields used in validation are pressure p, temperature T, density ρ, and velocity $\mathbf{U}$. Mach number is read from `Ma` where present; otherwise it is reconstructed from

$$M = \frac{|\mathbf{U}|}{\sqrt{\gamma R T}}, \qquad \gamma = 1.4, \quad R = 2.87 \times 10^2\,\mathrm{J\,kg^{-1}\,K^{-1}}. \tag{2}$$

All results reported here come from existing completed outputs. No solver rerun, mesh regeneration, cleaning, or time-directory deletion was performed for this report update.

5 Pressure-Ratio Study

Three pressure-ratio cases isolate the expected subsonic, choked, and internal-shock regimes. The latest-time validation metrics are shown in Table 2.

The subsonic case remains below Mach 1. The choked case reaches sonic throat conditions and accelerates downstream, but the existing output also shows internal-shock-like behavior and should not be interpreted as a clean shock-free isentropic expansion. The internal-shock case reaches sonic throat conditions, accelerates supersonically, and contains a shock-like downstream transition.

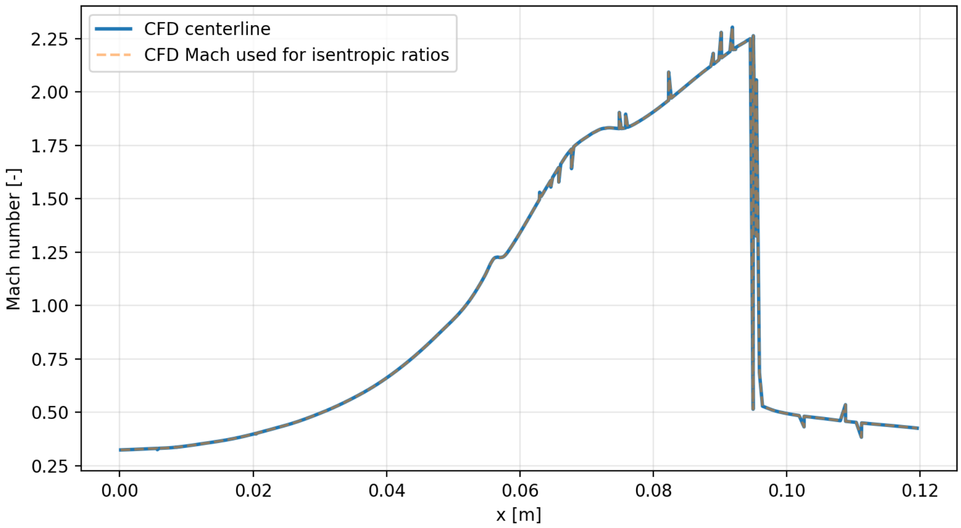

Figure 2: Choked-case centerline Mach number from the latest computed field.

6 Validation Methodology

Validation is performed from OpenFOAM ASCII outputs using deterministic Python post-processing. The workflow checks:

- latest numerical time and solver completion status;
- mesh quality from existing `checkMesh` logs;
- maximum and final Courant numbers from solver logs;
- min/max pressure, temperature, density, and Mach number;
- direct inlet/outlet mass-flow balance;
- throat Mach number, maximum Mach number, and observed flow regime;
- area-Mach agreement against quasi-1D theory;
- time-history behavior from logs and written fields.

The isentropic reference calculations use the local Mach number and ideal-gas air constants $\gamma = 1.4$ and $R = 2.87 \times 10^2\,\mathrm{J\,kg^{-1}\,K^{-1}}$. If a saved `Ma` field is absent, Mach number is reconstructed as

$$M = \frac{|\mathbf{U}|}{\sqrt{\gamma R T}}. \tag{3}$$

The static-to-stagnation ratios used for centerline pressure, temperature, and density diagnostics are

$$\frac{T}{T_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1}, \tag{4}$$

$$\frac{p}{p_0} = \left(\frac{T}{T_0}\right)^{\frac{\gamma}{\gamma-1}}, \tag{5}$$

$$\frac{\rho}{\rho_0} = \left(\frac{T}{T_0}\right)^{\frac{1}{\gamma-1}}. \tag{6}$$

Table 3: Area-Mach validation summary.

Case	RMS M error	Valid points	Masked points	Shock x [m]
`subsonic`	0.197192	459	0	–
`choked`	0.569107	459	0	–
`internal_shock`	0.0757175	370	89	0.0561075

Table 4: Time-history assessment from existing logs and saved fields.

Case	Log samples	Field samples	Max Co	Final mass error	Steadiness
`subsonic`	106396	11	0.612799	0.690756%	nearly steady
`choked`	58939	3	0.355479	0.277965%	quasi-steady
`internal_shock`	104026	5	0.201246	0.256732%	quasi-steady

The quasi-1D area-Mach model is evaluated on the appropriate branch using

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}. \tag{7}$$

This equation is applied only in regions where an inviscid isentropic comparison is meaningful; detected shock regions are masked because entropy production changes the downstream effective A^*.

Mass flow is computed from saved fields and mesh geometry:

$$\dot{m} = \int_A \rho \mathbf{U} \cdot \mathbf{n}\, dA. \tag{8}$$

The mass conservation error is computed from the difference between the absolute inlet and outlet mass-flow rates, normalized by the larger magnitude. The validation scripts do not use `phi`, `rhoPhi`, or ρ`phi`.

7 Area-Mach Validation

The area-Mach validation extracts centerline Mach data, reconstructs $A(x)/A^*$ from the nozzle geometry, and solves both branches of the isentropic relation. The internal-shock case masks the detected shock region and does not apply isentropic theory through the shock.

The choked case shows the largest area-Mach RMS difference, indicating that the computed flow is not a purely shock-free isentropic expansion over the whole divergent section. This is reported as a limitation of the case label rather than hidden. For the internal-shock case, masking the shock and post-shock entropy-change region gives a much lower pre-shock comparison error and a documented downstream fitted branch.

8 Time-History Assessment

Solver logs are parsed for time, Δt, Courant number, and execution time. Field-based histories are recomputed only at saved time directories using existing ρ, $\mathbf{U}$, T, and mesh geometry. Therefore, mass-flow and throat-Mach histories have lower temporal resolution than the solver log.

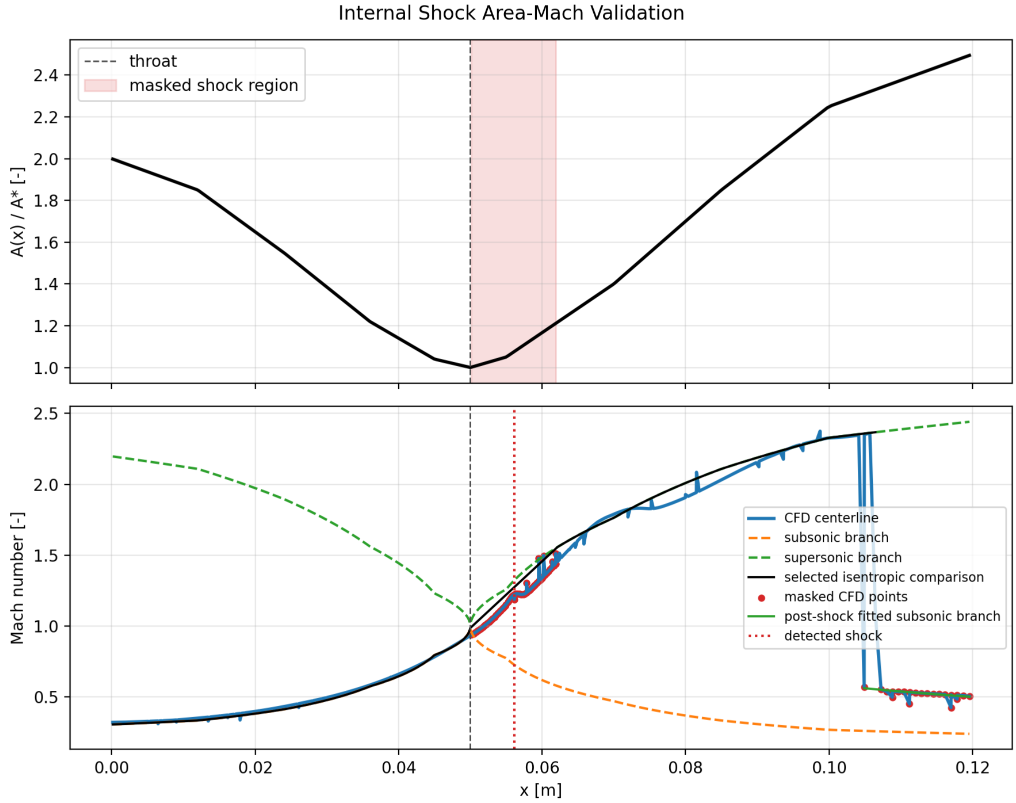

Figure 3: Internal-shock area-Mach validation. The shock region is masked and excluded from the isentropic comparison.

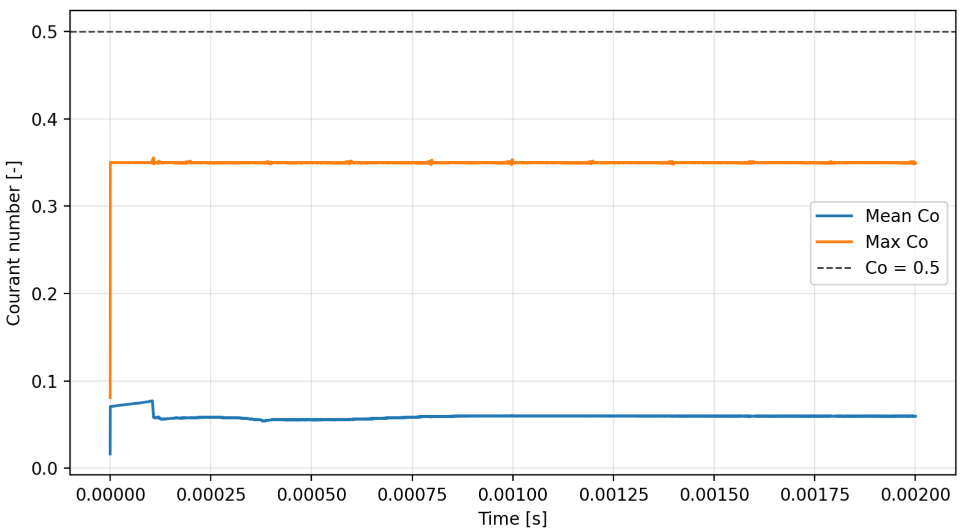

Figure 4: Choked-case Courant-number history parsed from the existing solver log.

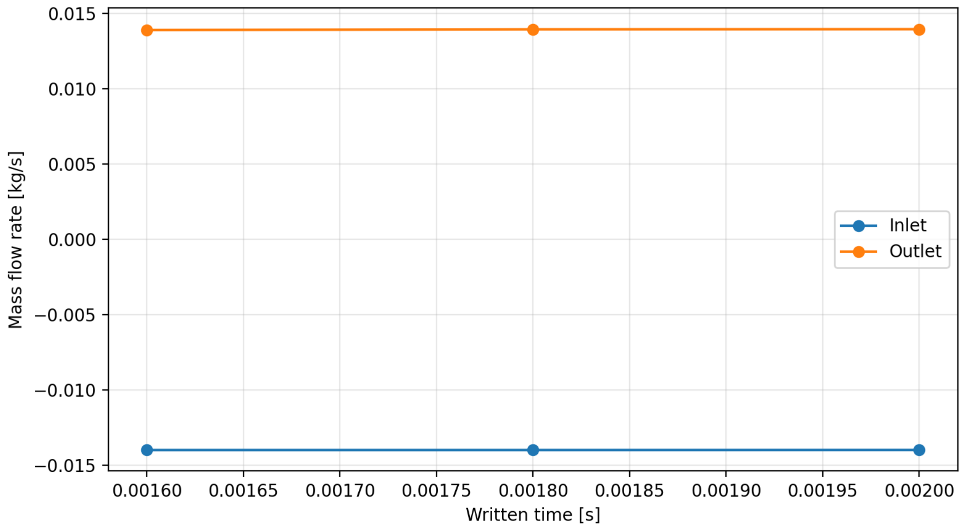

Figure 5: Choked-case inlet and outlet mass-flow history recomputed from written fields.

Table 5: Mesh independence summary for the choked operating point.

Mesh	Cells	Runtime [s]	Throat M	Max M	Mass error
`coarse`	20000	463.85	1.04648	2.2641	0.512397%
`medium`	70000	3500.52	1.03652	2.30388	0.277965%
`fine`	150000	12863.8	1.0468	2.2675	0.183476%

9 Mesh Independence Study

The mesh study varies only the `blockMeshDict` resolution for the choked operating point. Runtime is parsed from existing solver logs.

The medium-to-fine throat Mach change is 0.0102739, equal to 0.9815% relative to the fine result. The medium mesh is sufficient for regime classification and validation-level conclusions. For strict quantitative throat-Mach reporting with an absolute tolerance of 0.01, the fine mesh should be used or the residual medium-to-fine difference should be stated.

10 Results and Discussion

The current script suite supports complete validation of existing OpenFOAM outputs without requiring a solver rerun. It parses solver logs, mesh files, field data, centerline profiles, and written time directories to regenerate Markdown summaries, CSV tables, and plots for regime classification, mass balance, area-Mach behavior, and time histories.

The pressure-ratio study reproduces the expected qualitative sequence of Laval-nozzle regimes: fully subsonic flow, sonic throat with downstream acceleration, and shock-containing divergent-section flow. The existing choked output also has internal-shock-like behavior, so the directory name is retained as the intended pressure-ratio case while the observed behavior is stated explicitly. All primary cases have positive primitive fields, acceptable mass conservation, passing mesh checks, and stable Courant histories.

The direct mass-flow calculation is a central validation feature. It uses the saved density and velocity fields on inlet and outlet patches, avoiding dependence on flux fields that may not be written or may represent solver-specific intermediate quantities. The final mass conservation errors are below 1% for the three pressure-ratio cases.

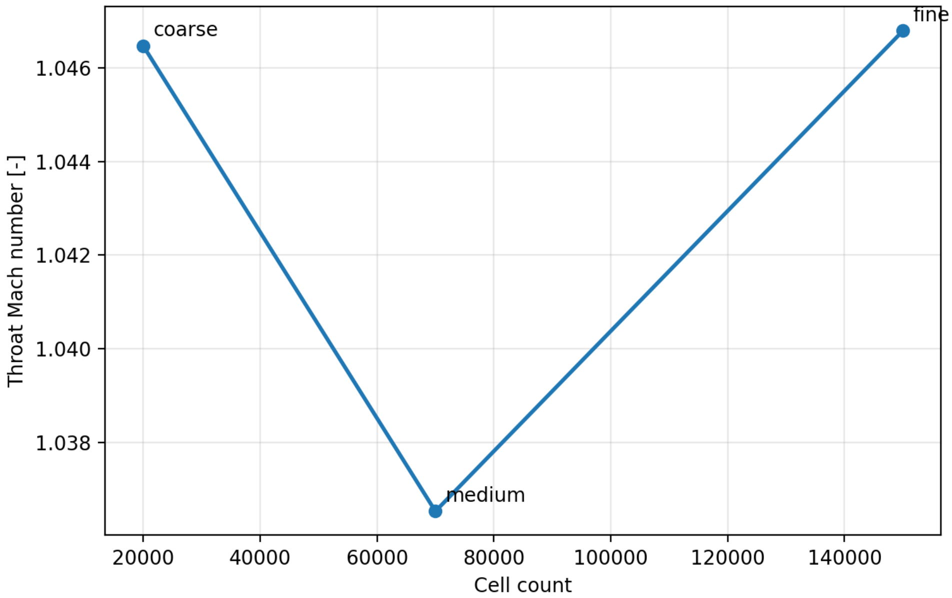

Figure 6: Throat Mach sensitivity across the mesh study.

The area-Mach comparison is useful as a diagnostic rather than a complete truth model. The internal-shock case is treated carefully by masking the detected shock region, while the higher RMS error in the choked case is reported rather than hidden. This distinction is important because geometric-throat isentropic theory is not valid through shock losses or non-isentropic adjustment regions.

11 Limitations

- The nozzle is quasi-2D with `empty` front/back patches, not a full 3D geometry.
- Slip walls are intentional for inviscid/isentropic validation; viscous losses, wall heat transfer, and boundary layers are not modeled.
- Turbulence modeling is not the focus of this project.
- Shock position and strength are sensitive to mesh resolution, numerical scheme, and back pressure.
- Field-derived histories are limited by the saved write intervals.
- The area-Mach relation is inviscid and isentropic; it is not applied through the detected shock region.
- This report is a compact portfolio validation artifact, not a certified nozzle design analysis.

12 Conclusions

The completed simulations demonstrate a reproducible OpenFOAM workflow for compressible nozzle validation. The computed pressure-ratio cases show the expected regime progression, and the direct $\rho \mathbf{U} \cdot \mathbf{n}$ mass-flow integration gives sub-1% final mass conservation error for the three primary cases. Area-Mach and time-history diagnostics provide additional evidence of physical consistency while documenting where ideal theory is not applicable. The mesh study supports the 70000-cell medium mesh for validation-level regime classification, with the fine mesh preferred for stricter quantitative throat-Mach reporting.

13 Reproducibility

The repository contains the OpenFOAM cases, Python validation scripts, Markdown summaries, CSV data tables, generated figures, and this report. The principal generated artifacts are:

- `docs/validation_summary.md`
- `docs/pressure_ratio_study.md`
- `docs/mesh_independence.md`
- `docs/area_mach_validation.md`
- `docs/time_history_assessment.md`
- `docs/data/area_mach_validation_summary.csv`
- `docs/data/time_history_summary.csv`
- `docs/data/mesh_independence_summary.csv`

Existing outputs can be revalidated without rerunning the solver using:

```
python3 scripts/validate_completed_cases.py
python3 scripts/advanced_validation.py
```

The PDF can be rebuilt from the repository root with:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error laval_nozzle_report.tex
```